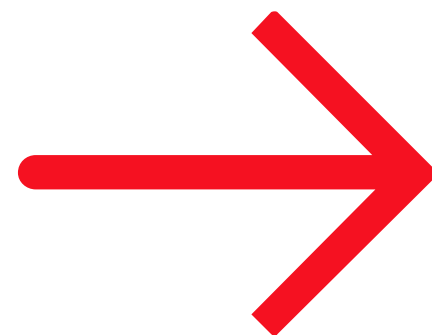




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OBSERVABILITY IN

AI



HOW TO DEBUG RAG & AGENTS PROPERLY



TRACES 01

02 **METRICS**

**EVALUATION
& FEEDBACK** 04

03 **LOGS**

**THE
4
CORE
PILLARS**

What is Observability?

Observability =

Your ability to look inside an AI system and understand what is happening.

Not just:

“The answer is wrong.”

But:

- Where did it fail?
- Retrieval?
- Prompt?
- Model?
- Tool call?

If you can't see inside, you're guessing.

Why It Matters in RAG

A RAG system is not just:

User → Retrieve → Generate

It's actually:

User

- Query rewrite
- Embedding
- Vector search
- Reranking
- Context building
- Prompt
- LLM

Failure can happen at ANY step.

Observability lets you trace it.

Observability in AI Agents

Agents are harder.

One user request can become:

User

- Agent reasoning
- Tool call
- Tool output
- Another tool
- Final answer

You must log:

- Each tool call
- Tool inputs & outputs
- Agent decisions

Otherwise debugging agents becomes chaos.

The 4 Pillars of AI Observability

Traces

Metrics

Logs

Feedback & Evaluation

These four together make your AI debuggable.

Let's break them down.

Traces (Full Story of One Request)

For every user query, you should see:

- Original query
- Rewritten query
- Retrieved chunks + scores
- Final context
- Prompt sent to LLM
- Model output

If the answer is wrong,
traces show you exactly where it broke.
This is the most important pillar for RAG.

Metrics (System Health)

Metrics tell you how your AI behaves at scale.

Track things like:

- Retrieval hit rate
- Similarity scores
- Token usage
- Latency per step
- Hallucination rate
- “No answer” rate

If similarity scores drop,
your knowledge base may be outdated.

Metrics = long-term visibility.

Logs (Engineering Layer)

Logs capture raw events:

- Vector DB timeout
- LLM failure
- Tool crash
- Empty response

Sometimes the issue isn't intelligence.
It's infrastructure.

Logs help you catch that.

Evaluation & Feedback(The Improvement Engine)

Evaluation & Feedback answer one big question:

Is the system actually getting better?

You need 3 layers here:

User Feedback

- 👍 / 👎 buttons
- “Was this helpful?”
- Report incorrect answers

Real users expose real failures.

Automated Evaluation

Use programmatic checks like:

- Faithfulness (Did the answer stick to context?)
- Relevance (Did it answer the question?)
- Context precision
- Groundedness

This helps you evaluate at scale without manual checking.

Benchmark / Test Sets

Create a fixed set of:

- Questions
- Expected answers
- Gold documents

Every time you change:

- Chunking
- Embeddings
- Prompt
- Model

Run the test set again.

If scores drop, you broke something.

What Happens Without Observability?

Without it:

You blame the LLM.

You tweak prompts randomly.

You over-engineer.

With observability:

You see:

- Wrong chunk retrieved
- Right chunk dropped
- Context too long
- Tool gave bad data

Now you fix the real issue.

Production AI is not about fancy prompts.

It's about:

Visibility.

Traceability.

Measurability.

If you cannot observe your AI,
you cannot improve it.

Build smart.

Observe everything.